Zunzhi You

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Education

The University of Sydney — Sydney, Australia
Ph.D. in Computer Science — *Mar. 2023 –*

Sun Yat-sen University — Guangzhou, China
B.E. in Software Engineering, School of Computer Science and Engineering — *Aug. 2017 – June 2021*
Overall Average: 90.25%, Ranking: 13/174, Outstanding Graduate

National Chiao Tung University — Hsinchu, Taiwan
Exchange Student in the Department of Computer Science — *Sep. 2019 – Jan. 2020*

Publications

Zunzhi You, Daochang Liu, Bohyung Han, Chang Xu. Beyond Pretrained Features: Noisy Image Modeling Provides Adversarial Defense. NeurIPS, 2023. [Paper] [Code]

Zunzhi You, Yi-Hsuan Tsai, Wei-Chen Chiu, Guanbin Li. Towards Interpretable Deep Networks for Monocular Depth Estimation. ICCV, 2021. [Paper] [Code]

Guanbin Li, Ricong Huang, Haofeng Li, Zunzhi You, Weikai Chen. SENSE: Self-Evolving Learning for Self-Supervised Monocular Depth Estimation. IEEE Transactions on Image Processing, 2023. [Paper]

Chung-Sheng Lai, Zunzhi You, Ching-Chun Huang, Yi-Hsuan Tsai, Wei-Chen Chiu. Colorization of Depth Map via Disentanglement. ECCV, 2020. [Paper] [Code]

Research Highlights

Sim2Real RL for Consecutive Rapid Turning with Legged Robot — *May 2024 – Sep 2025*
In submission to ICRA 2026. Advisors: Dr Chang Xu, Dr Yunke Wang
- Proposed to formulate the sequential local navigation task with a novel segment-based goal representation
- Developed an RL framework with a novel reward and look-ahead mechanism that learns an agile locomotion policy performing consecutive rapid turning with emergent anticipatory behaviors
- Deployed the learned policy on Unitree Go2 with ROS 2 and validated its robust sim-to-real transfer

Adversarial Defense from Noisy Image Modeling Pretraining — *Aug. 2022 – May 2023*
Advisors: Dr Chang Xu, Dr Daochang Liu, Professor Bohyung Han
- Implemented a self-supervised pretraining framework that brings adversarial robustness to downstream models
- Recognized that replacing masking in the mask image modeling framework by adding noise can learn strong noise-invariant features
- Exploited the reconstruction ability of the NIM model from noisy images to remove adversarial perturbations
- Showed that the proposed method improves the adversarial robustness with little sacrifice of the clean accuracy

Interpretability of DNNs for Monocular Depth Estimation — *Apr. 2020 – Mar. 2021*
Advisors: Dr Wei-Chen Chiu, Dr Yi-Hsuan Tsai, Dr Guanbin Li
- Observed some neural units in DNNs for monocular depth estimation are selective to certain depth ranges and more meaningful to the estimation performance
- Proposed to assign a depth range for each unit to select to and trained more interpretable and accurate monocular depth estimation model
- Validated the proposed method's reliability and applicability, e.g. providing cues to explain models' mistakes

Technical Skills

Hardware Platforms: Unitree Go2/A1, Jetson Orin Nano, RealSense D435i
Software Libraries: Isaac Lab, Isaac Gym, RSL RL, ROS 2, ROS, PyTorch
Programming Languages: Python, C++, Bash
Developer Tools: Git, Docker, Claude Code